

MARY MERZA

905-242-4032 | mariammerza@trentu.ca | <https://linkedin.com/in/marymerza> | <https://github.com/mary-merza>

SKILLS

Technical Skills: Python, SQL, pandas, scikit-learn, TensorFlow, Power BI, Excel, NumPy, machine learning, NLP, sentiment analysis, forecasting, data visualization, classification, regression, feature engineering, API integration, Docker, Git.

Languages: English (fluent).

EDUCATION

Bachelor of Science (Honours), Computer Science – Trent University | June 2026 | Oshawa, ON

PUBLICATION

Co-author, “Reducing Financial Debt and Illiteracy in Canadian Populations Using Machine Learning Prediction Models,” IEEE SNPD 2025.

EXPERIENCE

AI Program Researcher/Developer | Aristos Education Services Inc. | Jun 2026 - Present | Remote

- Designing and developing an AI educational program and curriculum for children and youth.
- Providing research and development services, putting together researched materials, sending correspondence in the form of updates, and creating plans or outlines for related tasks.

Data Research Analyst | Trent University | Sep 2025 – Jan 2026 | Oshawa, ON

- Built automated Python-based data collection pipelines integrating YouTube and Meta APIs to analyze 500k+ social media records.
- Applied NLP sentiment analysis and classification models to identify behavioural and engagement trends across large-scale COVID-19 datasets.
- Presented analytical findings and vaccine sentiment data insights to academic stakeholders through structured reporting and data visualization.

Undergraduate Research Fellow | Trent University | May 2025 – Jul 2025 | Oshawa, ON

- Developed ARIMA, LSTM, and hybrid forecasting models to analyze Canadian consumer financial data and predict credit-risk trends.
- Reduced forecasting error by 94% versus baseline ARIMA models using feature engineering and time-series optimization techniques.
- Presented research findings at the IEEE SNPD 2025 international conference to an audience of 200+ researchers.

PROJECTS

Fake News / Misinformation Detector | Python, scikit-learn, NLTK, APIs

Link: https://github.com/mary-merza/covid_misinformation_classifier

- Built a multi-class NLP classifier to detect fake news, misleading claims and factual reporting. Benchmarked 4 ML models using stratified 5-fold cross-validation; Logistic Regression achieved the best macro-F1 of 0.68.
- Designed for extensibility with real-world datasets (LIAR, FakeNewsNet).

Financial Forecasting & Recommendation System (ARIMA-LSTM Hybrid) | TensorFlow/Keras, pandas, NumPy, Matplotlib, Jupyter Notebook

Link: https://github.com/mary-merza/ARIMA_LSTM_financial_prediction

- Developed and trained a hybrid ARIMA-LSTM time-series model to forecast monthly household expenses, implemented anomaly detection (Isolation Forest) to flag unusual spending patterns and integrated a rule-based recommendation system to generate personalized financial advice.